Basics in Mathematical Physics

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1 Complex Analysis

1.1 Analytic Functions

An analytic function is a complex function that is differentiable at every point in its domain. If a function is differentiable within a region, then we say its analytic throughout the region. There are three nice properties of analytic functions:

1. Analytic functions are infinitely differentiable.
2. The contour integral of an analytic function over a closed curve is zero (Cauchy's theorem).
3. Analytic continuiation: If two analytic functions agree on a set of point that has an accumulation point, then they are equal on the entire domain.

Another condition for a function to be analytic is that it must satisfy the Cauchy-Riemann equations. If $f(z) = u(x, y) + iv(x, y)$, where $z = x + iy$, then the Cauchy-Riemann equations are given by:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{1}$$

1.2 Cauchy's Integral Theorem

Theorem 1.1 (Cauchy's Integral Theorem)**.** *If f is analytic in a simply connected domain D, then for any closed contour C in D, we have:*

$$\oint_C f(z)\,dz = 0. \tag{2}$$

Proof. We first expand $f(z)$ in terms of its real and imaginary parts, then the contour integral can be expressed as:

$$\oint_C f(z)\,dz = \oint_C (u + iv)(dx + idy) = \oint_C (udx - vdy) + i \oint_C (vdx + udy). \tag{3}$$

By Green's theorem, we can convert the line integrals into double integrals over the region R enclosed by C:

$$\oint_C (udx - vdy) = \iint_R \left(\frac{\partial(-v)}{\partial x} - \frac{\partial u}{\partial y}\right) dA, \quad \oint_C (vdx + udy) = \iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dA. \tag{4}$$

Since f is analytic, it satisfies the Cauchy-Riemann equations,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{5}$$

Substituting these into the double integrals, we find that both integrals evaluate to zero:

$$\oint_C (udx - vdy) = 0, \quad \oint_C (vdx + udy) = 0. \tag{6}$$

Therefore, we conclude that:

$$\oint_C f(z)\,dz = 0. \tag{7}$$

□

1.3 Cauchy's Integral Theorem

Theorem 1.2 (Cauchy's Integral Formula)**.** *If f is analytic in a simply connected domain D and C is a positively orirented simple closed contour in D, then for any point z_0 inside C, we have:*

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz. \tag{8}$$

Proof. For an arbitraryly small $\epsilon > 0$, we can find a δ such that for $|z - z_0| < \delta$

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \lambda(z)(z - z_0), \tag{9}$$

where $|\lambda(z)| < \epsilon$ (the definition of $f(z)$ being analytic at z_0). Deforming the contour to be the circle of radius δ about z_0 we find

$$\begin{aligned} \oint_C \frac{f(z)}{z - z_0} dz &= f(z_0) \oint_C \frac{1}{z - z_0} dz + f'(z_0) \oint_C dz + \oint_C \lambda(z) dz \\ &= i2\pi f(z_0) + \oint_C \lambda(z) dz \end{aligned} \tag{10}$$

As

$$\left| \oint_C \lambda(z) dz \right| \leq \oint_C |\lambda(z)||dz| < \epsilon 2\pi\delta, \tag{11}$$

we have

$$\left| \oint_C \frac{f(z)}{z - z_0} dz - 2\pi i f(z_0) \right| < \epsilon 2\pi\delta. \tag{12}$$

Since ϵ is arbitrarily small, we have

$$\oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0). \tag{13}$$

□

1.4 Derivatives from Cauchy's Integral Theorem

Takes derivative of the Cauchy's Integral Formula with respect to z_0, we have

$$f'(z_0) = \frac{1}{2\pi i} \frac{d}{dz_0} \oint_C \frac{f(z)}{z - z_0} dz = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz. \tag{14}$$

Then more genrally we will have the nth order derivative of f at z_0 as

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz. \tag{15}$$

1.5 Liouville's Theorem

Theorem 1.3 (Liouville's Theorem)**.** *If f is entire and bounded, then f is constant.*

Proof. If f is entire and bounded, then there exist a constant M such that $|f(z)| \leq M$ in the entire complex plane. By Cauchy's integral formula for the first derivative, we have:

$$f'(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z)^2} dw, \tag{16}$$

where C is a circle of radius R centered at z. Taking the modulus and using the bound on f, we get:

$$|f'(z)| \leq \frac{1}{2\pi} \oint_C \frac{|f(w)|}{|w - z|^2} |dw| \leq \frac{1}{2\pi} \cdot M \cdot \frac{2\pi R}{R^2} = \frac{M}{R}. \tag{17}$$

As $R \to \infty$, we have $|f'(z)| \to 0$, which means $f'(z) = 0$ for all z. Therefore, f is constant. □

1.6 Laurent Expansion

This is a very important theorem in complex analysis to express an analytic funcition in the form of power series. The Laurent expansion is a representation of a complex function as a series that includes both positive and negative powers of the variable. It is particularly useful for functions that have singularities, as it allows us to express the function in a neighborhood of those singularities.

Theorem 1.4 (Laurent Expansion)**.** *If f is analytic in an annulus $r < |z - z_0| < R$, then f has a unique representation as a Laurent series:*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \tag{18}$$

where the coefficients are given by:

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{n+1}} dw, \tag{19}$$

and C is any simple closed contour in the annulus, oriented positively.

If the function f is analytic throughout the whole disk, then the Laurent series reduces to a Taylor series. If f has a singularity at z_0, then the Laurent series will have negative powers of $(z - z_0)$. The principal part of the Laurent series is the sum of the terms with negative powers, and it captures the behavior of the function near the singularity.

Proof. If f is analytic in the annulus $r < |z - z_0| < R$, then the Cauchy integral formula gives us:

$$f(z) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w - z} dw - \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w - z} dw, \tag{20}$$

where C_1 is a positively oriented contour around z_0 with radius R, and C_2 is a negatively oriented contour around z_0 with radius r. We can expand the integrands in geometric series:

$$\frac{1}{w - z} = \frac{1}{w - z_0 - (z - z_0)} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(w - z_0)^{n+1}} \quad \text{for } |z - z_0| < |w - z_0|, \tag{21}$$

For the second term, we have:

$$-\frac{1}{w - z} = -\frac{1}{(z - z_0) - (w - z_0)} = \sum_{m=0}^{\infty} \frac{(w - z_0)^m}{(z - z_0)^{m+1}} \quad \text{for } |z - z_0| > |w - z_0|. \tag{22}$$

Then we substitue $n = -m - 1$ in the second series, we have

$$-\frac{1}{w - z} = \sum_{n=-\infty}^{-1} \frac{(w - z_0)^{-n-1}}{(z - z_0)^{-n}}. \tag{23}$$

Putting them together, we have

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n + \sum_{n=-\infty}^{-1} \left(\frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n. \tag{24}$$

Then this is the Laurent expansion of $f(z)$ in the annulus $r < |z - z_0| < R$,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \tag{25}$$

□

1.7 Isolated Singularities

Definition 1.1 (Isolated Singularity)**.** *A point $z_0 \in \mathbb{C}$ is called an isolated singularity of a complex function $f(z)$ if there exist $r > 0$ such that $f(z)$ is analytic in the punctured disk $0 < |z - z_0| < r$, but $f(z)$ is not analytic at z_0.*

Definition 1.2 (Removable Singularity)**.** *A removable singularity is an isolated singularity of a complex function $f(z)$ at a point z_0 such that the limit $\lim_{z\to z_0} f(z)$ exists and is finite. In this case, we can define a new function $\tilde{f}(z)$ that is equal to $f(z)$ for $z \neq z_0$ and is defined to be $\tilde{f}(z_0) = \lim_{z\to z_0} f(z)$. The new function $\tilde{f}(z)$ is then analytic at z_0, effectively "removing" the singularity.*

Definition 1.3 (Pole)**.** *A pole of order m is an isolated singularity of a complex function $f(z)$ at a point z_0 such that the Laurent series expansion of $f(z)$ around z_0 has $a_{-m} \neq 0$ and $a_{-n} = 0$ for $n > m$.*

In particular, the Laurent expansion for a order of m pole of complex function $f(z)$ in the region $0 < |z - z_0| < R$ is given by

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k + \frac{a_{-1}}{z - z_0} + \frac{a_{-2}}{(z - z_0)^2} + \cdots + \frac{a_{-m}}{(z - z_0)^m}. \tag{26}$$

Definition 1.4. *Essential Singularity An essential singularity is a type of isolated singularity of a complex function where the Laurent expansion of the function around that point contains infinitely many negative power terms.*

Then the Laurent expansion for a complex function $f(z)$ about its essential singularity is

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k + \sum_{n=-\infty}^{-1} a_n (z - z_0)^n, \quad a_n \neq 0 \text{ for infinitely many } n \tag{27}$$

2 Differential Equations